

Technical Design: Multi-Source Candidate Data Transformer

The system is a deterministic pipeline: parse source files, normalize extracted values, merge records into a canonical candidate, score confidence, project into the requested output shape, and validate before writing JSON. Parsers are intentionally thin adapters. Each parser emits `RawCandidateRecord`, so future sources such as GitHub, LinkedIn, ATS JSON, and recruiter notes can be added by implementing the same parser boundary without changing merge or projection logic.

The canonical schema is fixed internally and validated with Pydantic. It stores `candidate_id`, identity fields, contact arrays, normalized location, links, headline, `years_experience`, skills with confidence and sources, experience, education, field-level provenance, and `overall_confidence`. Phones normalize to E.164 with phonenumbers, countries normalize to ISO-3166 alpha-2 with `pycountry`, dates normalize to YYYY-MM, and skills map through a configurable alias dictionary.

Merge policy is deterministic and explainable. Emails and phones are normalized and deduplicated. Descriptive scalar fields prefer resume values over CSV values because resumes are candidate-authored and richer. Structured CSV still contributes corroboration and contact values. Experience is ordered by preferred source and deduplicated by company/title, which keeps the latest-looking resume entry ahead of CSV current-role hints. Every accepted or rejected extracted value creates provenance with source, method, value, and confidence.

Confidence is deliberately simple enough to defend in an interview. If multiple sources agree on a value, confidence is high. A single clean source is medium. Malformed values, such as invalid phone numbers, are recorded with low confidence and dropped from canonical contact arrays. Overall confidence is the average of non-zero field and item scores, making it stable and interpretable.

Projection is a separate runtime layer. The internal schema never changes. Config fields can select canonical paths, rename outputs, map array paths such as `skills[].name`, include or hide provenance and confidence, and choose a missing-value policy of null, omit, or error. Edge cases handled explicitly include missing files, empty PDFs, invalid CSVs, invalid phones, duplicate records, and missing projection paths. Under time pressure, I would leave out deep NLP resume parsing and external API connectors, but the adapter boundary is ready for them.